

Supplementary Material: Bosonic pair production and squeezing for optical phase measurements in long-lived dipoles coupled to a cavity

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PROPOSED EXPERIMENTAL IMPLEMENTATION

In the main text we introduced a proposal to realize an analog of bosonic four-wave mixing by coupling multilevel atoms to a far-detuned optical cavity mode. Here, we present a detailed discussion of the full atom-light Hamiltonian describing this system and show that it can be reduced to an effective spin model, which is presented as Eq. (1) in the main text.

Atom-light model

We consider multilevel atoms with a ground manifold and an excited manifold separated by an energy $\hbar\omega_a$, and $2F + 1$ Zeeman levels in each manifold. We denote the excited states as $|e, m\rangle$ and the ground states as $|g, m\rangle$ where the index $m \in [-F, F]$ specifies the Zeeman level. The cavity supports a pair of photon modes with degenerate angular frequency ω_c but different polarization. Without loss of generality, we take one of the cavity modes to be linearly polarized along the atomic quantization axis, which we call Π -polarization, and the other cavity mode to be linearly polarized perpendicular to the quantization axis, which we call Σ -polarization. The atoms couple to these two cavity modes with single-photon Rabi frequency $2g_0$.

The dynamics of the atom-light system is modeled by the Lindblad master equation,

$$\hbar \frac{d\rho}{dt} = -i[\hat{H}_{\text{tot}}, \rho] + \mathcal{L}_c[\rho] + \mathcal{L}_s[\rho]. \quad (\text{S.1})$$

Here, $\hat{H}_{\text{tot}} = \hat{H}_A + \hat{H}_L + \hat{H}_{AL}$ is a Hamiltonian including contributions from the atoms, cavity modes, and atom-light coupling:

$$\begin{aligned} \hat{H}_A &= \hbar\omega_a \hat{n}_e + \hbar(\delta_g \hat{F}_g^z + \delta_e \hat{F}_e^z), \\ \hat{H}_L &= \hbar\omega_c \sum_{\alpha=\Pi, \Sigma} \hat{c}_\alpha^\dagger \hat{c}_\alpha, \\ \hat{H}_{AL} &= \hbar g_0 (\hat{c}_\Pi \hat{\Pi}^+ + \hat{c}_\Sigma \hat{\Sigma}^+ + \text{h.c.}). \end{aligned} \quad (\text{S.2})$$

In the above equations, $\hat{n}_e$ is the occupation in the excited manifold, $\hat{F}_{g(e)}^z =$

$\sum_{i=1}^N \sum_{m=-F}^F m |g(e), m\rangle_i \langle g(e), m|$ is the azimuthal spin operator for the ground (excited) manifold, and $\hat{c}_\alpha$ annihilates a photon in the cavity mode with polarization α . The terms $\hbar\delta_g \hat{F}_g^z$ and $\hbar\delta_e \hat{F}_e^z$ arise from Zeeman shifts due to an applied magnetic field. The operators $\hat{\Sigma}^+(\hat{\Sigma}^-)$ and $\hat{\Pi}^+(\hat{\Pi}^-)$ are collective atomic operators that excite (de-excite) atoms by absorbing a Σ -polarized and Π -polarized cavity photon, respectively, given by $\hat{\Pi}^+ = \sum_{i,m} C_m^0 |e, m\rangle_i \langle g, m|$ and $\hat{\Sigma}^+ = i \sum_{i,m,q=\pm 1} C_m^q |e, m+q\rangle_i \langle g, m| / \sqrt{2}$, where $C_m^q = \langle F, m; 1, q | F, m+q \rangle$ is the Clebsch-Gordan coefficient associated with exciting from $|g, m\rangle$ to $|e, m+q\rangle$.

It is convenient to move to a rotating frame that rotates at the atomic frequency ω_a . In this frame, the atomic angular frequency and cavity frequency are shifted by ω_a , yielding the Hamiltonian

$$\begin{aligned} \hat{H}_{\text{tot}} &= \hbar(\delta_g \hat{F}_g^z + \delta_e \hat{F}_e^z) + \hbar\Delta \sum_{\alpha=\Pi, \Sigma} \hat{c}_\alpha^\dagger \hat{c}_\alpha \\ &+ \hbar g_0 (\hat{c}_\Pi \hat{\Pi}^+ + \hat{c}_\Sigma \hat{\Sigma}^+ + \text{h.c.}), \end{aligned} \quad (\text{S.3})$$

and we define the detuning of the cavity from the atomic transition, $\Delta = \omega_c - \omega_a$.

There are two Lindblad terms in the master equation (S.1), $\mathcal{L}_c[\rho]$ and $\mathcal{L}_s[\rho]$ that describe decoherence of the photon and atomic degrees of freedom. The former term captures leakage of photons out of the cavity at rate κ , and is given by

$$\mathcal{L}_c[\rho] = \hbar\kappa \sum_{\alpha=\Pi, \Sigma} \left(\hat{c}_\alpha \rho \hat{c}_\alpha^\dagger - \frac{1}{2} \hat{c}_\alpha^\dagger \hat{c}_\alpha \rho - \frac{1}{2} \rho \hat{c}_\alpha^\dagger \hat{c}_\alpha \right). \quad (\text{S.4})$$

The second Lindblad term, $\mathcal{L}_s[\rho]$, models spontaneous decay of atoms from the excited manifold at rate γ ,

$$\mathcal{L}_s[\rho] = \sum_{i=1}^N \sum_{l=\pi, \sigma_\pm} \left(\hat{L}_{l,i}^- \rho \hat{L}_{l,i}^+ - \frac{1}{2} \hat{L}_{l,i}^+ \hat{L}_{l,i}^- \rho - \frac{1}{2} \rho \hat{L}_{l,i}^+ \hat{L}_{l,i}^- \right). \quad (\text{S.5})$$

This expression is further decomposed into three kinds of Lindblad jump operators: $\hat{L}_{\pi,i}^- = \sqrt{\hbar\gamma} \sum_m C_m^0 |g, m\rangle_i \langle e, m|$ for spontaneous decay that preserves magnetization, and

$\hat{L}_{\sigma_{\pm},i}^- = \sqrt{\hbar\gamma} \sum_m C_m^{\pm 1} |g, m\rangle_i \langle e, m \pm 1|$ for spontaneous decay that leads to a change in magnetization by ± 1 .

Effective spin model

When the cavity is detuned sufficiently far from the atomic transition, $|\Delta| \gg g_0\sqrt{N}$, we can adiabatically eliminate the cavity photons and obtain an effective master equation for the atoms, $\hbar \frac{d\rho}{dt} = -i[\hat{H}, \rho] + \mathcal{L}_c[\rho] + \mathcal{L}_s[\rho]$ [1]. The effective Hamiltonian is

$$\hat{H} = \hbar\chi_0 (\hat{\Sigma}^+ \hat{\Sigma}^- + \hat{\Pi}^+ \hat{\Pi}^-) + \hbar(\delta_g \hat{F}_g^z + \delta_e \hat{F}_e^z), \quad (\text{S.6})$$

where $\chi_0 \simeq -g_0^2/\Delta$ and the Lindblad terms are

$$\begin{aligned} \mathcal{L}_c[\rho] = & \hbar\Gamma_0 \left(\hat{\Pi}^+ \rho \hat{\Pi}^- - \frac{1}{2} \hat{\Pi}^+ \hat{\Pi}^- \rho - \frac{1}{2} \rho \hat{\Pi}^+ \hat{\Pi}^- \right) \\ & + \hbar\Gamma_0 \left(\hat{\Sigma}^+ \rho \hat{\Sigma}^- - \frac{1}{2} \hat{\Sigma}^+ \hat{\Sigma}^- \rho - \frac{1}{2} \rho \hat{\Sigma}^+ \hat{\Sigma}^- \right), \\ \mathcal{L}_s[\rho] = & \sum_{i=1}^N \sum_{l=\pi, \sigma_{\pm}} \left(\hat{L}_{l,i}^- \rho \hat{L}_{l,i}^+ - \frac{1}{2} \hat{L}_{l,i}^+ \hat{L}_{l,i}^- \rho - \frac{1}{2} \rho \hat{L}_{l,i}^+ \hat{L}_{l,i}^- \right), \end{aligned} \quad (\text{S.7})$$

where $\Gamma_0 \approx g_0^2 \kappa / \Delta^2$. The terms in $\hat{H}$ proportional to χ_0 are cavity photon-mediated exchange of atomic excitations, and the terms in $\mathcal{L}_c[\rho]$ capture collective decay of atoms from the excited manifold, also called superradiant decay.

As discussed in the main text, we consider initial conditions where half of the atoms are in $|g, m = -F\rangle$ and half in $|e, m = F\rangle$. Thus of all the exchange processes mediated by the cavity photons and written in Eq. (S.6), the dominant exchange occurs in the $m = -F$ and $m = F$ sublevels, mediated by Π -polarized photons. The reason for this is that the Clebsch-Gordan coefficients are largest for these processes, $C_{\pm F}^0 = \pm \sqrt{F/(F+1)}$. The exchange process with the next highest amplitude, that exchanges atoms from $|g, -F\rangle |e, F\rangle$ to $|e, -F+1\rangle |g, F-1\rangle$ mediated by the exchange of Σ -polarized photon, has a smaller amplitude due to a smaller Clebsch-Gordan coefficient, $(C_F^{-1})^2 = 1/(F+1)$. Therefore, we expect that the populations in $|e, -F+1\rangle$ and $|g, F-1\rangle$ grow at an exponentially slower rate than in $|e, -F\rangle$ and $|g, F\rangle$. The amplitude of the process mediated by the Π -polarized photons, which exchanges atoms from $|g, -F\rangle |e, F\rangle$ to $|e, -F\rangle |g, F\rangle$, is proportional to $N\chi_0(C_F^0)^2$. At early times, the amplitudes of all the other exchange processes are at most $O(\sqrt{N}\chi_0)$.

Thus, keeping only the dominant exchange terms, from Eq. (S.6) we obtain the effective spin model [Eq. (1)] of the main text. The spin raising and lowering operators in Eq. (1) in the main text are the raising and lowering operators in $\Pi^{\pm}$ projected into the

A and B manifolds, i.e. $\hat{S}_A^+ = \sum_i |e, -F\rangle_i \langle g, -F|$, $\hat{S}_B^+ = -\sum_i |e, F\rangle_i \langle g, F|$, and the inversion operators are $\hat{S}_A^z = \sum_i (|e, -F\rangle_i \langle e, -F| - |g, -F\rangle_i \langle g, -F|)/2$ and $\hat{S}_B^z = \sum_i (|e, F\rangle_i \langle e, F| - |g, F\rangle_i \langle g, F|)/2$. The Clebsch-Gordan coefficients that appeared in $\hat{\Pi}^{\pm}$ now do not appear in the spin operators $\hat{S}_{A,B}^{\pm}$, but are instead included the definition of the parameters in Eq. (1) as $\chi \approx -g_F^2/\Delta$, where $g_F = g_0\sqrt{F/(F+1)}$, and $\delta = F(\delta_e - \delta_g)$.

To justify our argument that the effective spin model [Eq.(1) in the main text] is the full spin Hamiltonian [Eq. (S.6)] projected into the $m = \pm F$ manifolds, we demonstrate that a numerical simulation of the dynamics of the full spin model [Eq. (S.6)] with all $2F+2$ spin levels produces results consistent with an analytical prediction of the dynamics for the reduced effective spin model. In the next section, we describe how to analytically model the dynamics with an equivalent bosonic description of Eq.(1) and a subsequent approximation. In the section following that, we describe the numerical method that we use to simulate the full spin model [Eq.(1)].

PREPARING THE ATOMS IN TWO ENSEMBLES

Our protocol requires us to prepare $N/2$ atoms each in $|g, -F\rangle$ and $|e, F\rangle$. Previous experiments with ^{87}Sr have prepared atoms in $|g, m = \pm F\rangle$ using optical pumping [2, 3]. This was achieved via narrow-line laser cooling on the $^1S_0 \rightarrow ^3P_1$ transition, which is a common choice. If π -polarized light is used to interrogate the $^1S_0(F = 9/2) \rightarrow ^3P_1(F' = 7/2)$ transition, the $m_F = \pm 9/2$ states in the ground manifold are the dark states, and atoms will accumulate there at the end of cooling process. Due to symmetric Clebsch-Gordan coefficients, the populations in the ground $|m_F = \pm 9/2\rangle$ states are roughly $N/2$ each. This state preparation has been previously demonstrated in [3]. A final π pulse acting only on the $m_F = +9/2$ $^1S_0 \rightarrow ^3P_0$ transition, i.e. $|g, F\rangle \rightarrow |e, F\rangle$, using the differential g-factor between the $^1S_0 \rightarrow ^3P_0$ levels as demonstrated in Ref. [4], prepares the required initial state. This preparation protocol may lead to a statistical atom number imbalance between $|g, -F\rangle$ and $|e, F\rangle$ of the order of $\delta N \sim \sqrt{N}$. We later investigate the effects of this imbalance in more detail.

DYNAMICS IN EQUIVALENT BOSONIC DESCRIPTION

In the main text, we reported analytic expressions for the internal state occupations $n_1(t)$ and $n_2(t)$ based on the reduction of the multilevel system discussed in Sec. [and Eq. (1) of the main text] to an ideal bosonic two-mode squeezing process. Here, we derive these analytic expressions within UPA. The expressions in the main text

are for $\delta = N\chi/2$.

We start from the equivalent representation of Eq. (1) in the main text in terms of Schwinger bosons,

$$H = \hbar\chi(\hat{a}_{e,A}^\dagger\hat{a}_{g,A} + \hat{a}_{e,B}^\dagger\hat{a}_{g,B})(\hat{a}_{g,A}^\dagger\hat{a}_{e,A} + \hat{a}_{g,B}^\dagger\hat{a}_{e,B}) + \frac{\hbar\delta}{2}(\hat{a}_{e,B}^\dagger\hat{a}_{e,B} - \hat{a}_{g,B}^\dagger\hat{a}_{g,B} - \hat{a}_{e,A}^\dagger\hat{a}_{e,A} + \hat{a}_{g,A}^\dagger\hat{a}_{g,A}). \quad (\text{S.8})$$

Under the UPA, we replace $\hat{a}_{g,A}^\dagger\hat{a}_{g,A} = \frac{N}{2} - \hat{a}_{e,A}^\dagger\hat{a}_{e,A}$, $\hat{a}_{e,B}^\dagger\hat{a}_{e,B} = \frac{N}{2} - \hat{a}_{g,B}^\dagger\hat{a}_{g,B}$, and $\hat{a}_{g,A}^\dagger \approx \hat{a}_{e,B}^\dagger \approx \sqrt{\frac{N}{2}}$. Then, the Hamiltonian (S.8) reduces to

$$H \approx \frac{N\hbar\chi}{2}(\hat{a}_{e,A}^\dagger + \hat{a}_{g,B})(\hat{a}_{e,A} + \hat{a}_{g,B}^\dagger) - N\hbar\delta(\hat{a}_{e,A}^\dagger\hat{a}_{e,A} + \hat{a}_{g,B}^\dagger\hat{a}_{g,B}). \quad (\text{S.9})$$

Since the Hamiltonian (S.9) is quadratic in Schwinger boson operators, the Heisenberg time evolution of these operators can be analytically solved by integrating the matrix equation

$$i\partial_t \begin{pmatrix} \hat{a}_{e,A} \\ \hat{a}_{g,B}^\dagger \end{pmatrix} = \begin{pmatrix} \frac{N\chi}{2} - \delta & \frac{N\chi}{2} \\ -\frac{N\chi}{2} & -\frac{N\chi}{2} + \delta \end{pmatrix} \begin{pmatrix} \hat{a}_{e,A} \\ \hat{a}_{g,B}^\dagger \end{pmatrix}. \quad (\text{S.10})$$

The occupations are then obtained as,

$$n_{e,A}(t) = n_{g,B}(t) = \frac{(N\chi)^2}{4\delta(N\chi - \delta)} \sinh^2 \left(t\sqrt{\delta(N\chi - \delta)} \right). \quad (\text{S.11})$$

When $\delta = N\chi/2$ the Zeeman shift cancels the mean-field energy shift [corresponding to the elastic interaction terms in (S.8)], making the pair production resonant. Then, the occupations grow the fastest as $n_{e,A}(t) = n_{g,B}(t) = \sinh^2 \frac{N\chi t}{2}$.

NUMERICAL BENCHMARKS

Validity of the 4-level model approximation

In the main text, we approximated the full spin model [Eq. (S.6)] with a four-level spin model [Eq. (1)], based on the argument that the smaller amplitudes for scattering particles into Zeeman levels with $m \neq \pm F$ lead to negligible occupations in those levels (see also earlier discussion of the effective spin model). In this section, we verify the validity of this approximation, by comparing the dynamics produced by the full spin model involving many levels [Eq. (S.6)], with the dynamics produced by the reduced 4-level spin model [Eq. (1)]. The 4-level spin model is amenable to numerical exact diagonalization (ED) for moderate system sizes [$N \lesssim 10^5$] in the collective manifold of the two ensembles. The full spin model, which involves, e.g., 20 levels for ^{87}Sr , is not amenable to numerical exact diagonalization. Therefore we calculate the

dynamics in that case using a semiclassical method based on the truncated Wigner approximation (TWA).

To be concrete, in the TWA we obtain expectation values of observables by averaging over an ensemble of trajectories, and each trajectory is obtained by integrating classical equations of motion for initial values of operators sampled from a probability distribution which properly accounts for quantum fluctuations [5].

Specifically, we integrate classical equations of motion for collective multilevel spin operators $\hat{S}_{\alpha\beta} = \sum_i |\alpha\rangle_i \langle\beta|_i$, where α and β can be a ground state (g, m) or an excited state (e, m). There are $(4F+2)^2$ such spin operators. Their initial values are sampled from a multivariate Gaussian distribution with mean and covariance matrix given by the mean and covariance in the initial state,

$$\mu_{\alpha\beta} = \langle \hat{S}_{\alpha\beta} \rangle, \quad \sigma_{\alpha\beta, \gamma\delta} = \sqrt{\frac{1}{2} \langle \hat{S}_{\alpha\beta} \hat{S}_{\gamma\delta} + \hat{S}_{\gamma\delta} \hat{S}_{\alpha\beta} \rangle - \langle \hat{S}_{\alpha\beta} \rangle \langle \hat{S}_{\gamma\delta} \rangle}. \quad (\text{S.12})$$

After creating an ensemble of initial values $\{S_{\alpha\beta}(t=0)\}$, each initial value is then propagated according to their equations of motion (see below).

To obtain the equations of motion, it is convenient to first write the Hamiltonian in short-hand notation as

$$\hat{H} = \sum_{\alpha\beta} h_{\alpha\beta} \hat{S}_{\alpha\beta} + \sum_{\alpha\beta\gamma\delta} h_{\alpha\beta\gamma\delta} \hat{S}_{\alpha\beta} \hat{S}_{\gamma\delta}. \quad (\text{S.13})$$

Here, $h_{\alpha\beta}$ are single-particle terms due to the magnetic field, and $h_{\alpha\beta\gamma\delta}$ are interaction terms. For our system, $h_{\alpha\beta}$ is a diagonal matrix. The Heisenberg equations for the multilevel operators can be written schematically as

$$i\partial_t \hat{S}_{\alpha\beta}(t) = [\hat{S}_{\alpha\beta}, \hat{H}] = \sum_{\mu\nu} \Lambda_{\alpha\beta}^{\mu\nu} \hat{S}_{\mu\nu} + \sum_{\mu\nu\lambda\sigma} \Lambda_{\alpha\beta}^{\mu\nu\lambda\sigma} \hat{S}_{\mu\nu} \hat{S}_{\lambda\sigma} \quad (\text{S.14})$$

where $\Lambda_{\alpha\beta}^{\mu\nu}$ depends on the single-particle terms, and $\Lambda_{\alpha\beta}^{\mu\nu\lambda\sigma}$ depends on the interactions. To obtain the classical equations of motion, we replace quantum operators with their classical counterparts. Monomial operators are replaced as $\hat{S}_{\alpha\beta} \rightarrow S_{\alpha\beta}$, and quadratic operators are replaced by a symmetric decoupling scheme, $\frac{1}{2}\{\hat{S}_{\alpha\beta}, \hat{S}_{\gamma\delta}\} \rightarrow S_{\alpha\beta} S_{\gamma\delta}$. Subsequently, we integrate these equations using a standard differential equation solver.

Observables are computed by averaging over trajectories. For instance, occupations are obtained from $n_\alpha(t) = \overline{S_{\alpha\alpha}(t)}$, where $\overline{\dots}$ denotes ensemble average.

As a first comparison, in Fig. S.1 we compute the number of entangled pairs, $\bar{n}$, that populate the levels $|g, F\rangle$ and $|e, -F\rangle$. We find excellent agreement between TWA results, using a model with 20 levels, and a numerically exact calculation for the four-level model.

To complement this result, in Fig. S.2 we use TWA to demonstrate that leakage of population outside the

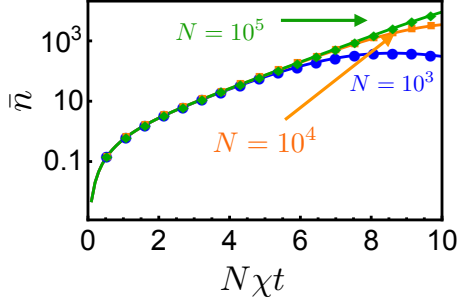


FIG. S.1. Growth of number of entangled particles $\bar{n}$ for $\delta = N\chi/2$. Results obtained from TWA treatment of 20-level spin model (dots) agree excellently with results from numerical exact diagonalization (solid lines) of the reduced 4-level spin model [Eq. (1) in the main text].

four-level model is negligible. Fig. S.2(a) plots the total occupation $\bar{n}$ in the hyperfine levels $|g, m \neq \pm F\rangle$ and $|e, m \neq \pm F\rangle$. This remains small and is relatively suppressed as time increases, validating our conjecture that the imbalance of the Clebsch-Gordan coefficients will lead to a runaway dominance of the quartet of levels $|g, \pm F\rangle$ and $|e, \pm F\rangle$. Moreover, Fig. S.2(b) plots the variance of the difference in population δn in $|e, -F\rangle$ and $|g, F\rangle$. Vanishing fluctuations in the population difference are a hallmark of the bosonic two-mode squeezing that is predicted for our simplified four-level model (see later discussion of UPA), whereas the involvement of the other 16 levels would lead to the introduction of excess population fluctuations. Our results in Fig. S2(b) demonstrate that these fluctuations remain small, relative to the level of uncorrelated Poissonian fluctuations $\sim \bar{n}$, throughout the timescales of interest for our protocol.

As a final subtle point, we note that the TWA method used in the treatment of the 20-level model was independently benchmarked by comparing a TWA treatment of the 4-level model to results generated by exact diagonalization. We found excellent agreement for all relevant variables across the timescales of interest in this proposal, which justifies our use of TWA for the full 20-level model, for which direct comparison to numerical exact diagonalization is not feasible for meaningful system sizes.

Validity of the UPA

In the main text, we approximated the 4-level spin model [Eq.(1)] to a bosonic pair production model [Eq. (3)] which is amenable to an analytic treatment when the UPA is implemented. We verify the UPA is applicable for short times in Fig. S.3 by comparing to a numerically exact calculation for moderate system sizes ($N < 10^5$). Specifically, we compute the number of entangled pairs $\bar{n}$ and the squeezing ξ^2 for fixed $\delta = N\chi/2$.

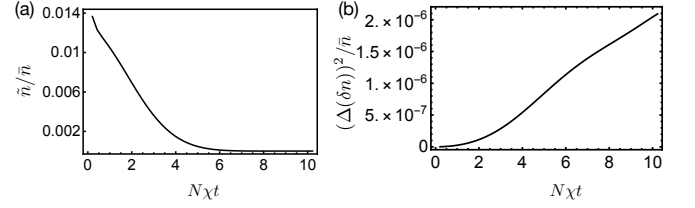


FIG. S.2. (a) Total occupation $\bar{n}$ of the hyperfine levels $|g, m \neq \pm F\rangle$ and $|e, m \neq \pm F\rangle$, normalized to the occupation $\bar{n}$ in $|g, F\rangle$ and $|e, -F\rangle$. (b) Variance $(\Delta(\delta n))^2$ of the occupation difference δn between $|g, F\rangle$ and $|e, -F\rangle$, normalized by $\bar{n}$. Both (a) and (b) are calculated using TWA for the full (20-level) spin model [Eq. (S.6)] and using fixed $\delta = N\chi/2$ and $N = 10^5$. Both $\bar{n}$ and $(\Delta(\delta n))^2$ remain vanishingly small compared to $\bar{n}$ throughout the dynamics (note vertical scales).

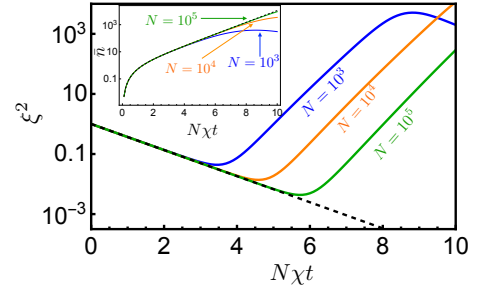


FIG. S.3. The squeezing ξ^2 , calculated using numerical exact diagonalization of Eq.(1) in the main text with fixed $\delta = N\chi/2$. Inset: Number of entangled pairs $\bar{n}$ versus time. Exact results agree with the UPA prediction (dashed lines) until finite-size effects (i.e., pump depletion) become important.

The results in Fig. S.3 show excellent agreement between the ED and UPA predictions at short timescales, as expected, before finite size effects (i.e, pump depletion) become relevant.

DESCRIPTION OF THE SQUEEZING IN JOINT BLOCH SPHERES

An important and well known feature of the bosonic model of pair production [Eqs. (S.8) and (S.9)] is that it generates squeezing in joint quadratures of the two bosonic modes. In this section, we show that this implies that, within UPA, our spin model is also capable of generating spin squeezing of combined spin quadratures of the A and B ensembles.

To be concrete, we consider only the case with $\delta = N\chi/2$ so that our effective spin Hamiltonian $\hat{H}$ [see Eq. (1) of the main text]] reduces to

$$\hat{H} = \hbar\chi(\hat{S}_A^+ \hat{S}_B^- + \hat{S}_B^+ \hat{S}_A^-). \quad (\text{S.15})$$

We begin by rewriting this Hamiltonian in a more convenient form,

$$\hat{H} = 2\hbar\chi(\hat{S}_A^x \hat{S}_B^x + \hat{S}_A^y \hat{S}_B^y), \quad (\text{S.16})$$

which can be alternatively factorized as,

$$\hat{H} = \hbar\chi \left(\hat{S}_B^y + \hat{S}_A^x \right) \left(\hat{S}_B^x + \hat{S}_A^y \right) - \hbar\chi \left(\hat{S}_B^y - \hat{S}_A^x \right) \left(\hat{S}_B^x - \hat{S}_A^y \right). \quad (\text{S.17})$$

To proceed, let us denote the terms appearing in Eq. (S.17) as $\mathcal{S}_{1,+} = \hat{S}_B^x + \hat{S}_A^y$, $\mathcal{S}_{2,+} = \hat{S}_B^y + \hat{S}_A^x$, $\mathcal{S}_{1,-} = \hat{S}_B^x - \hat{S}_A^y$, and $\mathcal{S}_{2,-} = \hat{S}_B^y - \hat{S}_A^x$. As we will see below, this new notation will enable us to gain physical intuition by visualizing the physics as occurring on two independent Bloch spheres. We define a third axis, $\mathcal{S}_{3,-} = \hat{S}_B^z - \hat{S}_A^z$, which enables us to identify that $(\mathcal{S}_{1,+}, \mathcal{S}_{2,+}, \mathcal{S}_{3,-})$ and $(\mathcal{S}_{1,-}, \mathcal{S}_{2,-}, \mathcal{S}_{3,-})$ each independently satisfy the canonical commutation relations of SU(2) spins. Therefore, we can construct two Bloch spheres with a shared axis, with the first sphere's axes as $(\mathcal{S}_{1,+}, \mathcal{S}_{2,+}, \mathcal{S}_{3,-})$ and the second sphere's axes as $(\mathcal{S}_{1,-}, \mathcal{S}_{2,-}, \mathcal{S}_{3,-})$. The first line of Eq. (S.17) acts only on the first Bloch sphere, and the second line acts only on the second Bloch sphere.

The generation of squeezing by $\hat{H}$ can be understood by mapping the axes of the Bloch spheres to bosonic quadratures, which translates $\hat{H}$ to a form that is familiar in quantum optics. At the mean-field level (equivalently, in the UPA), $\mathcal{S}_{3,-}$ is a constant of motion with value $N/2$. Therefore, in this approximation, $\mathcal{S}_{1,+}$ and $\mathcal{S}_{2,+}$ can be mapped to bosonic quadratures $\hat{X}_+$ and $\hat{Y}_+$ as $\mathcal{S}_{1,+} \approx \sqrt{\frac{N}{2}}\hat{X}_+$ and $\mathcal{S}_{2,+} \approx \sqrt{\frac{N}{2}}\hat{Y}_+$, which satisfy the canonical commutation relation $[\hat{X}_+, \hat{Y}_+] = i$. We similarly define $\mathcal{S}_{1,-} \approx \sqrt{\frac{N}{2}}\hat{X}_-$ and $\mathcal{S}_{2,-} \approx \sqrt{\frac{N}{2}}\hat{Y}_-$, where $[\hat{X}_-, \hat{Y}_-] = i$. The Hamiltonian, written in terms of these bosonic quadratures, is

$$\hat{H} = \frac{N\hbar\chi}{2}\hat{Y}_+\hat{X}_+ - \frac{N\hbar\chi}{2}\hat{Y}_-\hat{X}_-. \quad (\text{S.18})$$

It is well-known that this Hamiltonian generates squeezing along $\hat{Y}_+$ and $\hat{X}_-$ for $\chi > 0$ (and conversely, along $\hat{X}_+$ and $\hat{Y}_-$ for $\chi < 0$) [6]. Equivalently, we must have that our Hamiltonian generates squeezing in two *spin* quadratures along $\mathcal{S}_{2,+}$ and $\mathcal{S}_{1,-}$ respectively for $\chi > 0$. This is the *two-mode squeezing* that we report in the main text. The squeezing manifests itself as reduced quantum noise,

$$\text{var}(\mathcal{S}_{1,-}) = \text{var}(\mathcal{S}_{2,+}) = \frac{N}{4}e^{-\chi Nt}, \quad (\text{S.19})$$

and the anti-squeezing as increased quantum noise,

$$\text{var}(\mathcal{S}_{2,-}) = \text{var}(\mathcal{S}_{1,+}) = \frac{N}{4}e^{\chi Nt}. \quad (\text{S.20})$$

The squeezing/anti-squeezing is well-defined relative to the isotropic projection noise associated with a typical coherent spin state on each of the independent collective Bloch spheres, characterized by $\text{var}(\mathcal{S}_{1,\pm}) = \text{var}(\mathcal{S}_{2,\pm}) = N/4$ for our initial state polarized along $\mathcal{S}_{3,-}$.

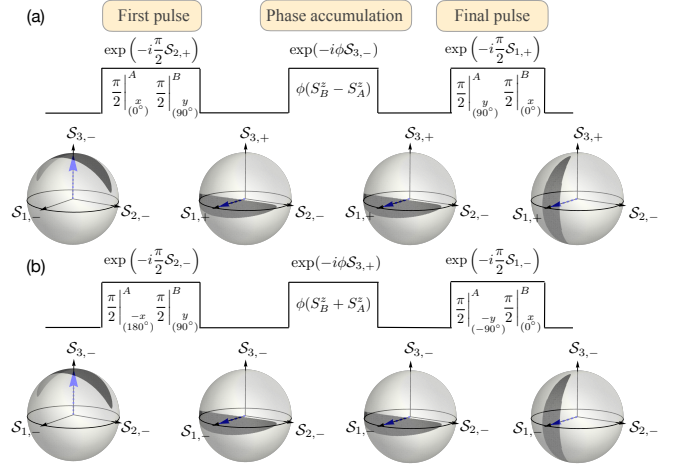


FIG. S.4. Evolution of the spin distributions in the upper Bloch sphere in Fig. 1(c) in the main text, during the Ramsey sequence for measuring (a) the differential phase, (b) the sum phase. Information about the imprinted phase is not captured in the spheres shown here. Dashed blue arrows show the Bloch vector.

Lastly, we briefly discuss the illustrations of the quantum noise in Fig. 1(c) of the main text. The dynamics, and in particular the squeezing, produced by the bosonic Hamiltonian $\hat{H} = \frac{N\hbar\chi}{2}\hat{Y}_+\hat{X}_+ - \frac{N\hbar\chi}{2}\hat{Y}_-\hat{X}_-$ can be understood via the Wigner function computed in terms of the bosonic quadratures. It is well known that the Wigner function of two-mode squeezed vacuum factorizes into a pair of single-mode squeezed states with respect to the independent phase-spaces defined by (X_+, Y_+) and (X_-, Y_-) . The axes describing these two phase spaces commute with each other. Strictly, when the UPA is satisfied we can transplant each of these bosonic phase-spaces (up to prefactors) to lie on the surface of the independent Bloch spheres defined by $(\mathcal{S}_{1,+}, \mathcal{S}_{2,+}, \mathcal{S}_{3,-})$ and $(\mathcal{S}_{1,-}, \mathcal{S}_{2,-}, \mathcal{S}_{3,-})$ by ignoring the curvature effects [i.e., the spin Wigner functions are effectively limited to the 2D planes $(\mathcal{S}_{1,+}, \mathcal{S}_{2,+})$ and $(\mathcal{S}_{1,-}, \mathcal{S}_{2,-})$, where the spin Wigner functions on each Bloch sphere lie on axes that commute with their counterparts on the other Bloch sphere, assuming the pump populations are equal]. We adopt this correspondence for Fig. 1(c) to schematically illustrate the spin-squeezing in this spirit. For illustrative purposes, we enlarge the area covered by the spin Wigner functions, while strictly within the UPA, they are restricted to an infinitesimal surface perpendicular to $\mathcal{S}_{3,-}$.

EVOLUTION OF THE JOINT BLOCH SPHERES DURING THE RAMSEY SEQUENCE

In the main text, we visualized the Ramsey sequence as rotations of the spin distribution on the lower Bloch

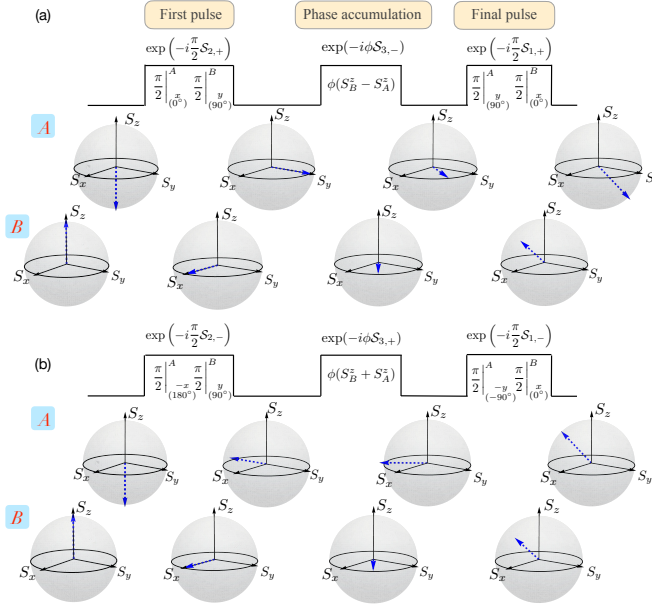


FIG. S.5. Rotation of the Bloch vectors in the A and B ensembles during the Ramsey sequence for measuring (a) the differential phase, (b) the sum phase.

sphere of Fig. 1(c). Here, we describe the evolution of the spin distribution on the upper Bloch sphere during the Ramsey sequence.

We first consider the Ramsey sequence that measures the differential phase [illustrated in Fig. 2(a) in the main text]. After the first pulse, realized by $\exp(-i\frac{\pi}{2}\hat{S}_{2,+})$, the axes on the upper Bloch sphere are rotated from $(\mathcal{S}_{1,-}, \mathcal{S}_{2,-}, \mathcal{S}_{3,-})$ to $(-\mathcal{S}_{3,+}, \mathcal{S}_{2,-}, \mathcal{S}_{1,+})$. This is illustrated in Fig. S.4(a). The Bloch vector points along $\mathcal{S}_{1,+}$, which is now a shared axis with the other Bloch sphere [see Fig. 2(a)]. At this step, the system is a null state of $\mathcal{S}_{1,-}$, assuming the initial pump populations were equal.

The next step in the Ramsey sequence is a small rotation ϕ of the Bloch vector. In the main text, we considered a rotation generated by $\exp(-i\phi\mathcal{S}_{3,-})$. This operation does not rotate the spin distribution in Fig. S.4(a). This is because in UPA, the spin distribution lies entirely in the plane $(\mathcal{S}_{3,+}, \mathcal{S}_{2,-})$, and these two axes commute with $\exp(-i\phi\mathcal{S}_{3,-})$, if the initial pump populations were equal. The rotation generated by $\exp(-i\phi\mathcal{S}_{3,-})$ only rotates the spin distribution in the Bloch sphere shown in the main text.

The final step in the Ramsey sequence is a rotation implemented by $\exp(-i\frac{\pi}{2}\mathcal{S}_{1,+})$, which rotates the spin distribution by 90° . Figure S.4(a) shows that the total atomic inversion encodes no information about ϕ , i.e. $\langle \hat{S}_A^z + \hat{S}_B^z \rangle = 0$. Conversely, Fig. 2(a) (main text) showed that the difference in atomic inversions, $\langle \hat{S}_B^z - \hat{S}_A^z \rangle$, encoded information about ϕ .

Fig. S.4(b) shows how the upper Bloch sphere in Fig. 1(c) evolves during the Ramsey protocol that mea-

sures the sum phase. No information about the imprinted sum phase is captured by the sphere in Fig. S.4(b).

For ease of understanding, we also show in Fig. S.5 how the Bloch vectors in the A and B ensembles separately evolve during the two Ramsey protocols.

ACHIEVABLE SENSITIVITY INCORPORATING INTRINSIC DECOHERENCE

In this section, we sketch how we compute the sensitivity of our protocol in the presence of decoherence. We begin by writing the master equation for $\hat{S}_A^\pm$ and $\hat{S}_B^\pm$, analogous to Eq. (S.10), but now including decoherence as well. The two sources of decoherence that we consider are collective decay and independent spontaneous decay of atoms from the excited states. These two decoherences have two main effects: they reduce coherences, and cause depletion of the excited modes. This will require us to go beyond the UPA, since the excited mode in the B ensemble is a pump mode, and it gets depleted by decay to the ground state. As a further effect of pump depletion, the Zeeman shift does not cancel the mean-field energy shift. We will account for this as well.

The master equation is

$$\begin{aligned} \partial_t \begin{pmatrix} \langle \hat{S}_A^- \rangle \\ \langle \hat{S}_B^- \rangle \end{pmatrix} &\approx \begin{pmatrix} -\frac{\gamma}{2} + i\delta & 0 \\ 0 & -\frac{\gamma}{2} - i\delta \end{pmatrix} \begin{pmatrix} \langle \hat{S}_A^- \rangle \\ \langle \hat{S}_B^- \rangle \end{pmatrix} \\ &+ (\Gamma + 2i\chi) \begin{pmatrix} \langle \hat{S}_A^z \rangle & \langle \hat{S}_A^z \rangle \\ \langle \hat{S}_B^z \rangle & \langle \hat{S}_B^z \rangle \end{pmatrix} \begin{pmatrix} \langle \hat{S}_A^- \rangle \\ \langle \hat{S}_B^- \rangle \end{pmatrix}, \end{aligned} \quad (\text{S.21})$$

where we have approximated that $\langle \hat{S}_A^z \hat{S}_B^- \rangle \approx \langle \hat{S}_A^z \rangle \langle \hat{S}_B^- \rangle$ and similarly for $A \leftrightarrow B$. Since spontaneous decay causes the excited populations to decrease with time, we have to modify the undepleted pump approximation, $\langle \hat{S}_B^z \rangle = \frac{N}{4}$, to $\langle \hat{S}_B^z \rangle = \frac{N}{4}(2e^{-\gamma t} - 1)$. We set $\langle \hat{S}_A^z \rangle = -N/4$, since there is no initial excited population in the A ensemble. We choose $\delta = \frac{N\chi}{2}$ as we did in the main text.

Equation (S.21) is a time-dependent coupled differential equation which is non-trivial to solve. Instead, we solve Eq. (S.21) to first order in Γ and γ . Separating the matrix on the right hand side as $M = M_0 + M_1(t)$, where M_0 is the contribution at $\Gamma = \gamma = 0$, and $M_1(t)$ is the remaining contribution, the solution to Eq. (S.21) is

$$\begin{aligned} \begin{pmatrix} \langle \hat{S}_A^-(t) \rangle \\ \langle \hat{S}_B^-(t) \rangle \end{pmatrix} &= \left(e^{M_0 t} + \int_0^t d\tau e^{M_0(t-\tau)} M_1(\tau) e^{M_0 \tau} \right) \\ &\times \begin{pmatrix} \langle \hat{S}_A^-(0) \rangle \\ \langle \hat{S}_B^-(0) \rangle \end{pmatrix}. \end{aligned} \quad (\text{S.22})$$

Similar equations can be set up and solved for the second moments, $\langle \hat{S}_A^+ \hat{S}_B^- \rangle$, $\langle \hat{S}_A^+ \hat{S}_A^- \rangle$, and $\langle \hat{S}_B^+ \hat{S}_B^- \rangle$, which are necessary for calculating the squeezed quantum noise.

In the Ramsey protocol that measures the differential phase, the value of the signal measured is

$$\begin{aligned} \langle \hat{S}_B^z - \hat{S}_A^z \rangle_{\text{final}} &= \langle \hat{S}_B^z(t) - \hat{S}_A^z(t) \rangle \sin \phi \\ &+ \langle \hat{S}_B^y(t) + \hat{S}_A^x(t) \rangle \cos \phi, \end{aligned} \quad (\text{S.23})$$

where t is the time before the first $\pi/2$ pulse in the Ramsey sequence. The first term in the above equation is $-\frac{N}{2}e^{-\gamma t} \sin \phi$, and the second term is zero. The fluctuation in the signal is

$$\begin{aligned} (\Delta(\hat{S}_B^z - \hat{S}_A^z))_{\text{final}}^2 &= (\Delta(\hat{S}_B^z(t) - \hat{S}_A^z(t)))^2 \sin^2 \phi \\ &+ (\Delta(\hat{S}_B^y(t) + \hat{S}_A^x(t)))^2 \cos^2 \phi \\ &+ \cos \phi \sin \phi \langle (\hat{S}_B^z(t) - \hat{S}_A^z(t))(\hat{S}_B^y(t) + \hat{S}_A^x(t)) \rangle \\ &+ \cos \phi \sin \phi \langle (\hat{S}_B^y(t) + \hat{S}_A^x(t))(\hat{S}_B^z(t) - \hat{S}_A^z(t)) \rangle. \end{aligned} \quad (\text{S.24})$$

The best sensitivity is achieved at $\phi = 0$, therefore we set $\phi = 0$ hereafter. Solving for the second order moments in a similar fashion to Eq. (S.22), the fluctuation in the signal is

$$\begin{aligned} (\Delta(\hat{S}_B^z - \hat{S}_A^z))_{\text{final}}^2 &= \frac{N}{4} \left(e^{-N\chi t} + \frac{\Gamma}{2\chi} + \frac{\gamma}{N\chi} \right. \\ &\left. + e^{N\chi t} \left(\frac{\Gamma}{4\chi} + \frac{\gamma}{2N\chi} - \frac{\gamma t}{2} \right)^2 + e^{N\chi t} \left(\frac{\gamma}{2N\chi} - \frac{\gamma t}{2} \right)^2 \right). \end{aligned} \quad (\text{S.25})$$

This yields the sensitivity as

$$\begin{aligned} (\Delta\phi)^2 &= \frac{e^{-N\chi t}}{N} + \frac{\Gamma}{2N\chi} + \frac{\gamma}{N^2\chi} + \frac{e^{N\chi t}}{N} \left(\frac{\gamma}{2N\chi} - \frac{\gamma t}{2} \right)^2 \\ &+ \frac{e^{N\chi t}}{N} \left(\frac{\Gamma}{4\chi} + \frac{\gamma}{2N\chi} - \frac{\gamma t}{2} \right)^2. \end{aligned} \quad (\text{S.26})$$

An analysis for the Ramsey protocol that measures the sum phase yields the same expression for the sensitivity.

SCALING OF $(\Delta\phi_{\text{min}})^2$ WITH N .

The best sensitivity is achieved at optimum values of Δ and t . These optimum values aim to find the right balance between the gain obtained by reaching a higher $\bar{n}$ versus the loss in squeezing due to decoherence. We find the optimum values of Δ and t by setting the derivatives of $\Delta\phi^2$ with respect to Δ and t as zero.

First, we find the optimum duration for our protocol. This is obtained by setting $d(\Delta\phi)^2/dt = 0$, which yields the implicit equation

$$\gamma t = \frac{\Gamma}{4\chi} + \sqrt{2e^{-2N\chi t} + \left(\frac{\gamma}{N\chi} \right)^2 - \left(\frac{\Gamma}{4\chi} \right)^2}. \quad (\text{S.27})$$

We approximate this time as $\gamma t \approx \frac{\Gamma}{4\chi} + \sqrt{2}e^{-N\Gamma/4\gamma}$. At this optimum time, the sensitivity is given by

$$\begin{aligned} N(\Delta\phi)^2 &= \exp \left(-\frac{N\Gamma}{4\gamma} - \frac{\sqrt{2}N\chi}{\gamma} e^{-N\Gamma/4\gamma} \right) + \frac{\Gamma}{2\chi} + \frac{\gamma}{N\chi} \\ &+ \exp \left(\frac{N\Gamma}{4\gamma} + \frac{\sqrt{2}N\chi}{\gamma} e^{-N\Gamma/4\gamma} \right) \\ &\times \left(\frac{\gamma^2}{N^2\chi^2} + e^{-N\Gamma/2\gamma} - \frac{\sqrt{2}\gamma}{N\chi} e^{-N\Gamma/4\gamma} \right). \end{aligned} \quad (\text{S.28})$$

The parameters Γ and χ depend on the cavity detuning Δ and the cavity loss rate κ as $\chi = \frac{g_F^2}{\Delta}$ and $\Gamma = \frac{g_F^2\kappa}{\Delta^2}$. Rewriting the sensitivity in terms of Δ , κ , and the cavity cooperativity $C = \frac{4g_F^2}{\kappa\gamma}$, we optimize the sensitivity with respect to Δ . This gives us an optimum value of Δ ,

$$\Delta \approx \sqrt{\frac{NC}{2\ln(2NC)}}\kappa. \quad (\text{S.29})$$

The optimum sensitivity at this detuning is given by

$$N(\Delta\phi)^2 \approx \sqrt{\frac{2\ln(2NC)}{NC}}. \quad (\text{S.30})$$

EFFECT OF PUMP POPULATION FLUCTUATIONS ON THE SENSITIVITY

Here, we calculate the sensitivity when the pump populations differ from the idealized case of $N/2$. The populations of the pump states $|g, A\rangle$ and $|e, B\rangle$ are parameterized as $N_{g,A} = (N + \delta N_{\text{tot}} + \delta N_{AB})/2$ and $N_{e,B} = (N + \delta N_{\text{tot}} - \delta N_{AB})/2$, where δN_{AB} and δN_{tot} characterize fluctuations of the difference and sum of the populations of the A and B ensembles, respectively.

The expression of the relevant spin variables in terms of the Schwinger bosons $\hat{a}_{g,B}$ and $\hat{a}_{e,A}$ is then,

$$\begin{aligned} \mathcal{S}_{1,\pm} &\approx \sqrt{N_{e,B}} \frac{\hat{a}_{g,B} + \hat{a}_{g,B}^\dagger}{2} \pm \sqrt{N_{g,A}} \frac{\hat{a}_{e,A}^\dagger - \hat{a}_{e,A}}{2i}, \\ \mathcal{S}_{2,\pm} &\approx \sqrt{N_{e,B}} \frac{\hat{a}_{g,B} - \hat{a}_{g,B}^\dagger}{2i} \pm \sqrt{N_{g,A}} \frac{\hat{a}_{e,A}^\dagger + \hat{a}_{e,A}}{2}, \\ \mathcal{S}_{3,-} &= \frac{N_{e,B} + N_{g,A}}{2} - \hat{a}_{e,A}^\dagger \hat{a}_{e,A} - \hat{a}_{g,B}^\dagger \hat{a}_{g,B} \approx \frac{N + \delta N_{\text{tot}}}{2}. \end{aligned} \quad (\text{S.31})$$

From these relations, we obtain the Heisenberg equation of motion for $\hat{a}_{e,A}$ and $\hat{a}_{g,B}$ within UPA as

$$i\partial_t \begin{pmatrix} \hat{a}_{e,A} \\ \hat{a}_{g,B}^\dagger \end{pmatrix} = \begin{pmatrix} N_{g,A}\chi - \delta & \sqrt{N_{g,A}N_{e,B}\chi} \\ -\sqrt{N_{g,A}N_{e,B}\chi} & -N_{e,B}\chi + \delta \end{pmatrix} \begin{pmatrix} \hat{a}_{e,A} \\ \hat{a}_{g,B}^\dagger \end{pmatrix}. \quad (\text{S.32})$$

These are solved analytically to obtain relevant expectation values, as per the ideal case previously discussed.

We focus on the effect of population fluctuations on the sensitivity of our Ramsey protocol, which for $\phi = 0$ is given by

$$(\Delta\phi)^2 = \frac{\text{var}(\mathcal{S}_{1,-}(t))}{\langle \mathcal{S}_{3,-}(t) \rangle^2}. \quad (\text{S.33})$$

Here, we have used that the relevant squeezed quadrature remains $\mathcal{S}_{1,-}$ even in the presence of small population fluctuations (which can be explicitly shown by diagonalizing the associated covariance matrix of the squeezed state).

Assuming that the sum and difference of the initial pump populations vary between shots of an experiment as a Gaussian random variable with zero mean and rms fluctuations σ_{AB} and σ_{tot} , respectively, the relevant expectation values generated by the solution of Eq. (S.32) are

$$\begin{aligned} \text{var}(\bar{\mathcal{S}}_{1,-}) &\approx \frac{N}{4} e^{-N\chi t} + \frac{\sigma_{\text{tot}}^2 + \sigma_{AB}^2}{16N} e^{N\chi t}, \\ \langle \bar{\mathcal{S}}_{3,-} \rangle &\approx \frac{N}{2}. \end{aligned} \quad (\text{S.34})$$

The approximate sign indicates that these results have been generated under the assumption that $\sigma_{AB}, \sigma_{\text{tot}} \ll N$. The sensitivity is thus

$$(\Delta\phi)^2 \approx \frac{e^{-N\chi t}}{N} + \frac{\sigma_{\text{tot}}^2 + \sigma_{AB}^2}{4N^3} e^{N\chi t}. \quad (\text{S.35})$$

Insight into this result can be obtained by recalling that finite size effects beyond UPA limit the achievable squeezing to $(\Delta\phi)^2 \sim N^{-3/2}$ and reached at timescales $e^{N\chi t} \sim \sqrt{N}$ (note that decoherence limits the sensitivity similarly). Making the replacement $e^{N\chi t} \rightarrow \sqrt{N}$ in the latter terms of Eq. (S.35) we thus determine that if the population fluctuations are sufficiently small, i.e., if $\sigma_{AB}, \sigma_{\text{tot}} \lesssim \sqrt{N}$, then the corrections to the sensitivity will also be $O(N^{-3/2})$. Phrased alternatively, our predicted sensitivity is robust to number fluctuations at the level of the shot noise.

PAIR PRODUCTION WITH SPIN-1 BECS

In the main text we contrast our effective bosonic model with alternative realization in quantum optics and spinor Bose-Einstein condensates. In the latter case, degenerate four-wave mixing is engineered through *s*-wave atomic collisions that change the internal Zeeman state of the atoms [7]. It is common to simplify the theoretical treatment of the atomic collisions by invoking a single mode approximation (SMA), wherein it is assumed that all atoms share the same spatial wavefunction, regardless of their internal Zeeman state. This is typically valid for small systems and short to intermediate timescales, although a range of factors contribute [8].

By freezing out the spatial degree of freedom via the SMA the internal state dynamics for a spin-1 BEC is then given by,

$$\hat{H}_{\text{BEC}} = \frac{U_s}{2N} \hat{\vec{S}} \cdot \hat{\vec{S}} + q(\hat{n}_1 + \hat{n}_{-1}). \quad (\text{S.36})$$

Here, the spin operators are $\hat{S}_x = (\hat{a}_1^\dagger \hat{a}_0 + \hat{a}_0^\dagger \hat{a}_{-1} + \text{h.c.})/\sqrt{2}$, $\hat{S}_y = (\hat{a}_1^\dagger \hat{a}_0 + \hat{a}_0^\dagger \hat{a}_{-1} - \text{h.c.})/\sqrt{2}i$, and $\hat{S}_z = \hat{a}_1^\dagger \hat{a}_1 - \hat{a}_{-1}^\dagger \hat{a}_{-1}$, where $\hat{a}_m^\dagger$ creates a boson in Zeeman sub-level $m = 0, \pm 1$. The first term of Eq. (S.36) describes spin-mixing due to *s*-wave atomic collisions characterized by interaction strength U_s , whereas the second term describes a quadratic Zeeman shift $q \propto B^2$ due to an applied magnetic field.

The Hamiltonian (S.36) can be rewritten in a more insightful form, $\hat{H}_{\text{BEC}} = \hat{H}_{\text{inel}} + \hat{H}_{\text{el}} + \hat{H}_Z$ with [9]

$$\begin{aligned} \hat{H}_{\text{inel}} &= \frac{U_s}{N} (\hat{a}_0 \hat{a}_0 \hat{a}_1^\dagger \hat{a}_{-1}^\dagger + \text{h.c.}), \\ \hat{H}_{\text{el}} &= \frac{U_s}{N} \hat{n}_0 (\hat{n}_1 + \hat{n}_{-1}) + \frac{U_s}{2N} (\hat{n}_1 - \hat{n}_{-1})^2, \\ \hat{H}_Z &= q(\hat{n}_1 + \hat{n}_{-1}). \end{aligned} \quad (\text{S.37})$$

The first term, $\hat{H}_{\text{inel}}$, describes spin-changing collisions wherein a pair of $m = 0$ atoms collide and scatter into an $m = \pm 1$ pair. Conversely, the second term, $\hat{H}_{\text{el}}$, describes elastic collisions which preserve the relative spin populations. Inspection of $\hat{H}_{\text{BEC}}$ in this form shows that, up to the distinguishing feature that the spin-1 BEC involves only three bosonic modes, it is analogous to the four-wave mixing Hamiltonian for our system described in Eq. (S.8).

Many experiments probe the regime where a BEC is prepared with the vast majority of atoms in the $m = 0$ state. For large systems the UPA then corresponds to replacing $\hat{a}_0 \approx \sqrt{N}$ where N is the number of atoms in the condensate. Equation (S.36) then reduces to

$$\hat{H}_{\text{BEC}} \approx (q + U_s)(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_{-1}^\dagger \hat{a}_{-1}) + U_s(\hat{a}_1^\dagger \hat{a}_{-1}^\dagger + \text{h.c.}). \quad (\text{S.38})$$

where we have ignored a term $\propto \hat{a}_1^\dagger \hat{a}_1 - \hat{a}_{-1}^\dagger \hat{a}_{-1}$ as a conserved quantity. Within the UPA, the spin-1 Hamiltonian is thus analogous to $\hat{H}_{\text{TMS}}$ in Eq. (3) in the main text, with $U_s = N\hbar\chi/2$, and $q = \hbar\delta$. The resonant condition for pair production is met when $q = -U_s$.

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